

D4 peak window LADESPERRE (0.48-c: pure function bm_d4)
goal: stacks stay empty in the morning, at midday they take the PV above the 20 kW export limit

clear-sky model dc(t) (bm_logic.c calc_arrays_kt, from fox2db_logic.h / clearsky.tex)

inputs	
time	unix clock (NTP, Fritzbox)
site	LAT 47.6811 N, LON 11.5732 E (Lenggries)
outdoor temperature	aussen/temp (MQTT, < 15 min old)
steps	
sun	NOAA: elevation, azimuth from time + site
atmosphere	air mass AM = 1 / sin(elev) (≤ 37), Meinel T = 0.7^(AM^0.678)
4 PV strings	tilt / azimuth from south / power, fitted per string (gen_pv_strings.py, 14 clear days): Sofar PV1 west 60 deg / +33 deg / 19430 W, PV2 south 68 deg / -12 deg / 7690 W FoxESS pv2 east 59 deg / -29 deg / 22036 W, pv1 west 67 deg / +32 deg / 2781 W
horizon	trees east: below 23 deg (south-east 15 deg) only 25 % diffuse
month	KT_MONTH empirical: Dec 0.13 ... May 0.91
temperature	cell = outdoor + NOCT 45 x sun; -0.30 %/K above 25 deg C (0.85..1.0)
dc(t) = Σ power x T x kt x cos(incidence) x shade x temperature factor	

model part (ladesperre wrapper, not a decision)
dc now: peak_h (hour 5..20 with max dc), win_end (last hour > 20 kW), solar noon
ratio = (dc - (PCC + Multis + Bat1)) / dc (5 min mean and now; dc > 5 kW)

bm_d4 bits	
new_day	local day changed: reset the state
enabled	BATMONITOR_LADESPERRE
pcc_peak	PCC fresh > 20 kW
season	May..August
day_can_peak	clear-sky max > 20 kW
before_peak	hour ≤ peak_h, before solar noon
ratio_valid / le_on / ge_off	5 min ratio valid, ≤ 0.50, ≥ 0.75
now_bad	ratio now ≥ 0.20

bm_d4 state: peak_today, latched, badweather
new_day -> all 0; pcc_peak -> peak_today
in_window = enabled and season and day_can_peak and before_peak and not peak_today
in_window and now_bad -> badweather (for the day)
latch: on at le_on, off at ge_off, off outside the window
block = in_window and latched and not badweather
[make check: 4096 combinations, 12 block; fixed point]

observation: only 12 of 4096 combinations block:
one cycle with ratio now ≥ 0.20 (house load counts)
ends the window for the day; NOW_RATIO 0.20 < latch 0.50

inputs
PCC (+ = export), PCC 5 min mean, Sofar Bat1 W (+ = charge), Multis AC-in,
outdoor temperature, BMS SoC / U / CCL, time (local, solar noon)

D5 DO4 pulse -> MQTT **pv_relay/DO4** (FoxESS shedding)
fresh PCC (< 15 s) > 20 kW -> "1", after 3 s "0", at most every 5 min
[make check]

surplus (chain_mode)
Bat1_eff = Bat1 discharge x 1, Bat1 charge x BAT1_CHARGE_FACTOR 0.5
surplus = z2 + Bat1_eff + own_charge - own_discharge
z2 = PCC + WP part the stacks do not cover (S2, as discharge)

D2 bm_d2 per stack state: prot, force, chg_prev
charge_mode: surplus > 200 W on, held while > 20 W
first match: BMS_MISSING > FORCE > STALE > LADESPERRE (charge_mode and block) >
CHARGE (SoC < 100) / FULL > PROT > CHARGING > DIS
[make check, CBMC h_d2_bits]

mode matrix BM_MODE_RULE[mode] (0.48-c): what each mode means for the setpoint

mode	action	capped	rule text
BMS_MISSING	0 W	-	NAME
FORCE	FORCE_CHARGE_W 500	1 (0.48-c)	FORCE(soc %)
STALE	0 W	-	NAME
LADESPERRE	0 W	-	LADESPERRE(peak h)
CHARGE	charge amount	1	CHARGE WP: SRC:
FULL	0 W	-	NAME
PROT	0 W	-	PROT(soc %)
CHARGING	0 W	-	CHARGING(BMS W)
DIS	discharge amount	1	(S4: PROP / IDLE)

chg_cap per phase (one per stack)
min(CHARGER_A 70 A x U / 0.93, CHARGE_MAX_PHASE 4900, BMS CCL)
~4.0-4.2 kW per phase -> ~12 kW for the three Multis

FORCE: min(FORCE_CHARGE_W, chg_cap) per phase
from the grid, obeys the BMS CCL (was: 500 W even at CCL 0)

charge amount (chain_charge, whole watts since 0.48-c)
want = max(0, own_charge + KP_PM 1010 x (z2 + Bat1_eff - SOYO_TARGET) / 1000)
at most the sum of chg_cap over the charging phases
[CBMC h_charge]

alloc_levels (0.48-c: one split core for charge and discharge)
charge: one level per stack in CHARGE_PRIORITY order (Stack1, Stack2), the lead stack last,
each up to min(chg_cap, CHARGER_CAP_PHASE); water filling within a level;
only what the caps cut goes on; a rest up to CHARGE_MAX_PHASE on all charging phases
[CBMC h_charge without bank assumption, h_alloc_discharge]

setpoints per phase (+ = charge) -> Multis
rule text: CHARGE WP:... SRC:... BAL/EQ [CCL] | LADESPERRE(h) | FULL | FORCE(soc)

clear summer day: how the window keeps the PCC below 20 kW			
time	PV / PCC	batmonitor	stacks
05:00 .. peak	PV rising, export < 20 kW	LADESPERRE (block)	0 W, stay empty
export would pass 20 kW	PCC > 20 kW once	peak_today -> window ends -> CHARGE; DO4 pulse sheds the FoxESS	take up to ~12 kW
peak hour / solar noon	highest PV	window over anyway -> CHARGE	export ~12 kW
afternoon	PV falling	CHARGE until FULL	-> 100 %
cloudy morning	shortfall ≥ 20 %	badweather -> no block	charge at once